

Project Report - Student Performance Predictor

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Course: Introduction to Problem Solving and Programming

1. Project Title

Student Performance Predictor

2. Objective

To design a simple machine learning model that predicts whether a student will Pass or Fail based on their marks, attendance, and average daily study hours using basic Python and scikit-learn.

3. Introduction

The project applies Logistic Regression to predict student performance. It uses a synthetic dataset generated within the code, making it fully self-contained and easy to run.

4. Modules Used

pandas - data handling

scikit-learn - model training & evaluation

random - dataset generation

5. Methodology / Algorithm

1. Generate random dataset of marks, attendance, study hours.
2. Label data (1 = Pass, 0 = Fail) based on weighted score.
3. Split data into training/testing sets.
4. Train Logistic Regression model.
5. Evaluate accuracy and predict Pass/Fail.

6. Tools & Technologies

Python 3, pandas, scikit-learn, Visual Studio Code

7. Model Evaluation

Test Samples: 300

Accuracy: 0.8667

Classification Report:

Class | Precision | Recall | F1-Score | Support

0 (Fail) | 0.4000 | 0.0513 | 0.0909 | 39

1 (Pass) | 0.8746 | 0.9885 | 0.9281 | 261

Accuracy: 0.8667 (300 samples)

Macro Avg: 0.6373 | 0.5199 | 0.5095

Weighted Avg: 0.8129 | 0.8667 | 0.8192

8. Example Data (First 5 Rows)

marks | attendance | study_hours | pass

48.72 | 95.58 | 4.10 | 1

79.96 | 96.42 | 6.22 | 1

69.24 | 74.73 | 7.35 | 1

42.41 | 77.55 | 6.74 | 1

56.32 | 63.36 | 8.81 | 1

9. Result

The model achieved 86.67% accuracy. It performs well in predicting student outcomes with consistent reliability for basic prediction tasks.

10. Conclusion

This project demonstrates basic machine learning concepts and classification using simple Python modules. It provides a foundation for further development using real datasets and visualization.